

Project Proposal: Library Management System

Course Code: CSE 282 Section: 4

Department of Computer Science and Engineering

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1. Team Members

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2. Project Overview

The Library Management System is a Java-based desktop application designed to provide automated stock tracking, full CRUD record management, persistent file storage, and structured reporting. Developed to meet all requirements of the Complex Engineering Project, the system allows warehouse and retail administrators to register new product entities, adjust stock levels, execute updates, remove records, and view transaction logs seamlessly through a Java Swing frontend.

The primary objective of this project is to apply all four core Object-Oriented Programming (OOP) concepts in Java specifically **Encapsulation, Abstraction, Inheritance, and Polymorphism** alongside robust custom exception handling and file-based data persistence across a clean 4-member team module split.

3. Key Features

1. Register New Entity (Create): Add new books and member records with fields for ID, name, author/category, and relevant details.
2. Display & View Records (Read): Fetch and view active book and member lists in Java Swing data tables with real time availability status.
3. Modify Library Details (Update): Update book information, member details, or availability status with instant UI updates.
4. Remove Library Record (Delete): Safely remove discontinued book or member records from active lists with

confirmation dialogues.

5. Search & Filter Module: Filter books by ID, title, author, category, or availability status.

6. File-Based Data Persistence: Automatically save library and transaction records to flat file storage (txt), ensuring

data reloads across system restarts.

7. Structured Exception Handling: Handle duplicate IDs, unavailable books, missing files, invalid data, and type mismatches using custom exceptions and try-catch blocks.

8. Java Swing Multi-Panel GUI: A structured frontend featuring distinct navigation panels for book management, member management, transaction logging, and summary reports.

9. Summary & Library Reporting: Generate automated text summary reports showing total books, available books,

issued books, member records, and transaction logs.

10. Business Rule Enforcement: Enforce strict library constraints such as preventing issue of unavailable books or

accepting invalid member/book data

4. Implementation of OOP Principles

To fully satisfy Task 1 of the Complex Engineering Project guidelines, the software architecture directly integrates all

four core OOP pillars:

- Encapsulation: Core entity attributes (such as bookId, title, author, and availability) are declared as private.

Input

validation logic is embedded within public setters to protect data integrity.

- Abstraction: An abstract base class LibraryItem provides common field definitions while declaring an abstract method public abstract String getItemDetails();.

- Inheritance: Concrete classes Book and ReferenceBook extend LibraryItem, reusing core methods while adding specific variables (e.g., author or reference type).

- Polymorphism: Both subclasses override the abstract method getItemDetails():

Book: Provides regular book information and issue/return details.

ReferenceBook: Provides reference-specific information and library usage rules..

5. Class Structure and Architecture

Class Name	Type	Variables / Fields	Methods / Responsibilities
Category	Model	categoryId, categoryName	Getters, setters, getCategoryDetails().
Library Item	Abstract class	Item id , name, category Availability	Getters/setters, abstract getItemDetails().

Book	Subclass	Author issuestatus	Overrides getItemDetails() with book and issue logic.
Reference Book	Subclass	Reference type	Overrides getItemDetails() with reference-book logic.

6. Team Work Distribution

The tasks are distributed equally among five team members according to individual project modules:

Member	Module / Deliverable	OOP / Java Focus
Member 1	OOP Architecture + Book Management	LibraryEntity.java, Book.java, ReferenceBook.java, BookManager.java
Member 2	Member Management + File Persistence	Member.java, MemberManager.java, FileManager.java
Member 3	Issue & Return Management	IssueRecord.java, IssueManager.java
Member 4	Exceptions + Reports	Exception classes, ReportManager.java
Member 5	Swing GUI + Integration	All GUI classes, Main.java

7. Conclusion

This project strikes an optimal balance between academic rigor and practical engineering requirements. By structuring clear OOP class hierarchies, full file-based CRUD operations, custom exception handling, and an intuitive Java Swing GUI, our team of five will deliver a robust, functional library management system in strict accordance with the CSE282 course specifications.